

1. Describe the workflow for SBDD.
2. Write a note on tools and resources used in SBDD.
3. Explain the purpose of Molecular Docking, Molecular mechanics and Molecular dynamics.
4. Write a short note on Scoring Functions.
5. Explain Energy minimization.
6. Explain the types of ligand-protein interactions.
7. Write a short note on history and evolution of SBDD.
8. Explain Free-Wilson Analysis in QSAR and its advantages over Hansch Analysis.
9. Explain ideal features of descriptors for the development of the QSAR models.
10. What are the essential steps involved in designing a QSAR experiment?
11. Discuss the use of genetic algorithm in QSAR modeling.
12. Write a note on validation process in QSAR
13. Discuss basic chemistry involved in QSAR.
14. Discuss applications of QSAR.
15. Explain Randomization.
16. Write a note on 2D QSAR models.
17. Discuss applications of all the types of QSAR methods (1D to 6D).
18. What are the key statistical parameters used to evaluate QSAR models?
19. Differentiate between QSAR, QSPR and QSTR.
20. Explain the use of genetic algorithms in QSAR modeling.
21. Compare CoMFA, CoMSIA, CoRIA and AFMoC methods in QSAR.
22. Describe the steps involved in homology modeling of protein structures.
23. Write a short note on Energy minimization techniques.
24. Explain the process of molecular docking.
25. Write a short note on Molecular mechanics.
26. What are the applications of Molecular dynamics.
27. Explain the difference between SBDD and LBDD.
28. Compare the Conformation Generation Methods for Molecular Docking
29. What are the essential steps involved in designing a QSAR experiment?
30. Discuss early QSAR methods in detail.
31. Explain key features of QSAR.
32. Describe the process to derive QSAR equations.
33. Discuss the types of Machine Learning models used in QSAR.
34. Discuss the advantages of random forest algorithm over linear regression models.
35. Differentiate between SAR and QSAR with examples.
36. Discuss the role of bootstrapping and randomization in QSAR model validation.
37. Explain 3D QSAR methods.
38. Define  $R^2$ ,  $Q^2$ , and RMSE.
39. Define all types of QSAR descriptors with examples (1D to 6D).

40. Discuss the importance of descriptor selection in QSAR modeling. What methods can be used to select relevant descriptors?
41. Compare CoMFA, CoMSIA, CoRIA and AFMoC methods in QSAR.
42. Describe LFER descriptors and their role in QSAR modelling.
43. Explain the process of energy minimization.
44. Compare the techniques in Molecular Modelling
45. Explain the concept of scoring functions in molecular docking.
46. Describe different non-bonded interactions in molecular mechanics.
47. Discuss the applications of Molecular Docking
48. What is Molecular modelling and its importance?
49. Describe the workflow of SBDD.
50. Describe the historical development of QSAR, emphasising the contributions of Corwin Hansch and Free-Wilson.
51. Differentiate between SAR and QSAR with examples.
52. Define QSAR and explain its significance in drug discovery and chemical research.
53. What are the key statistical parameters used to evaluate QSAR models?
54. Explain how overfitting affects QSAR models and how it can be prevented.
55. Describe the difference between a training set and a test set in QSAR modeling. Why is it important to split the dataset?
56. Explain the concept of molecular descriptors used in QSAR modeling with examples.
57. Explain the importance of Applicability domains.
58. Write a note on Bootstrapping.
59. Write a note 3D QSAR Methods.
60. Explain Randomization.
61. Compare all the types of QSAR.
62. What are the main differences between 3D QSAR and higher-order QSAR methods?
63. Why is validation important in QSAR.